

Rahul Manocha

5+ years in Data Science

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🌐 Web Portfolio

📄 PROFILE SUMMARY

AI Engineer | Data Engineer | LLM & Automation Specialist with 5+ years of experience in building scalable data pipelines, web scraping systems, and AI-driven solutions including Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs). Expertise in Python, ETL workflows, and machine learning, delivering high-impact automation and data-driven insights across e-commerce, analytics, and business intelligence domains.

📁 PROFESSIONAL EXPERIENCE

Jan 2025 – present
Gurgaon

Alternative Path Pvt. Ltd.

Web Scraping - Data Engineer

- Designed and developed scalable web scraping pipelines to extract structured and unstructured data from multiple domains.
- Built end-to-end ETL pipelines ensuring high data quality, validation, and reliability.
- Automated data workflows using orchestration tools, improving pipeline efficiency and reducing manual effort.
- Implemented monitoring and validation mechanisms to maintain data accuracy and consistency.

Feb 2023 – Jan 2025
Mohali

Kinerk Direct Marketing Solutions

Software Developer (Promoted as SD Data Science)

- Developed and deployed RAG pipelines to enhance information retrieval from large datasets, improving response accuracy.
- Integrated and optimized open-source LLMs for domain-specific applications, balancing performance and efficiency.
- Designed scalable data pipelines for web scraping across multiple e-commerce platforms.

Apr 2021 – Feb 2023
Mohali

Kinerk Direct Marketing Solutions

Web Admin (Data Analyst)

- Processed and transformed raw data into actionable insights using dashboards and reports.
- Designed visually compelling email templates using HTML/CSS to improve engagement.
- Bridged the gap between technical data and business insights through effective visualization.

📁 PROJECTS

AI-Powered GitHub PR Review Agent

Developed an AI-powered agent to automate pull request reviews by analyzing code changes and generating actionable feedback directly on PRs. Integrated the solution with GitHub Actions to trigger automated reviews on each submission, ensuring continuous code quality checks. Designed a scoring mechanism to evaluate PR quality, enabling automatic approval for high-scoring changes while escalating lower-scoring PRs to senior developers for further review. This system significantly improved review efficiency, reduced manual effort, and maintained consistent code quality standards.

Web Data Extraction (Multi-Domain)

Led the development of scalable web scraping systems delivering over 200 data products across finance, automotive, and market intelligence domains. Built robust scrapers capable of handling dynamic websites, anti-scraping mechanisms, and complex data structures, including index methodologies and multi-country heavy-duty truck registration datasets. Ensured high data accuracy, reliability, and timely delivery by implementing validation pipelines, monitoring systems, and structured code review practices.

Gardening Planner using Generative AI

Developed a Gardening Planner using Generative AI (Llama 3) and an RAG pipeline to provide personalized gardening recommendations. The planner customizes plant suggestions, care tips, and pesticide recommendations based on user inputs like soil type, budget, and hardiness zone.

Google Analytics 4 Data Fetch via API

Developed a Flask-based web application to retrieve and display Google Analytics 4 data via API integration. Enabled seamless data access and visualization through an intuitive HTML front-end interface.

Customer Segmentation using Machine Learning

Implemented customer segmentation using historical data by performing **RFM analysis** and **K-Means clustering**. Identified and segmented potential customers to enable targeted marketing strategies.

E-Commerce Reviews Analyzing System

Developed an E-Commerce Reviews Analyzing System with a Streamlit-based interactive dashboard to visualize product and category review trends. Conducted **sentiment analysis using VADER (NLP)** and generated word clouds to identify key customer concerns and preferences.

EDUCATION

Jun 2023	Masters in Data Science <i>Chandigarh University</i>
Jun 2019	Bachelor of Computer Application <i>Mr. Govt College (Panjab University)</i>

SKILLS

- **Languages:** Python, SQL
- **Frameworks & Libraries:** Flask, Pandas, NumPy, Streamlit
- **Machine Learning & AI:** Scikit-learn, TensorFlow, Keras, PyTorch, NLP, Generative AI, LLMs, RAG, Agentic AI, Fine-Tuning
- **Data Engineering:** ETL Pipelines, Apache Airflow (DAGs, Scheduling, Monitoring)
- **Databases & Warehousing:** MySQL, PostgreSQL, SQL Server, SSIS, SSMS, NoSQL, Snowflake
- **DevOps & Tools:** Docker, CI/CD Pipelines, GitHub Actions
- **Web Scraping:** Selenium, Scrapy, Playwright, BeautifulSoup, Requests
- **Data Visualization:** Power BI, Tableau, Matplotlib, Seaborn

CERTIFICATES

Google Data Analytics Professional Certificate | 2022